Synthesis of Reversible Circuits for Testing with Universal Test Set and C-Testability of Reversible Iterative Logic Arrays

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ABSTRACT

Reversibility is of interest in the design of very low-power circuits; it is essential for quantum computation. This paper examines the testability of an important subclass of reversible logic circuits that are composed of k-wire controlled NOT (k-CNOT) gates. Most commonly used stuck-at fault model (both single stuck-at fault (i.e. SSF) and multiple stuck-at fault (i.e. MSF)) has been assumed to be type of fault for such circuits. We define a universal test set (UTS) for a family C(n) of n-input circuits with respect to fault model F as a family of test sets T_{UTS} such that each C(n) has a unique test set T(n) in T_{UTS} that detects all F-type faults in every member of C(n). We show that if $k \geq 2$ for all gates, then the n-wire reversible circuits have a UTS of size n with respect to MSFs. By synthesizing 0-CNOT (inverters) and 1-CNOT gates from 2-CNOT (Toffoli) gates this result can be extended to all circuits of interest. We also present a method for modifying an n-wire reversible circuit to reduce its UTS size to 3. By modeling a k-CNOT gate as a k-input AND gate and a 2-input EXOR gate we then examine testability for the SSF model. Noting their resemblance to classical (irreversible) Reed-Muller circuits, which are well known to be easily testable, we prove that the n-wire reversible circuits have a UTS of size $n^2 + 2n + 2$. Finally, we turn to the reversible counterparts of another easily-testable classical circuit family, iterative logic arrays (ILAs). We define d-dimensional reversible ILAs (RILAs) and prove that they require a constant number test vectors irrespective of array length under the single cell fault (i.e. SCF) model; this number is determined by the size of the RILA cell's state table.

Index Terms— *Design for test, Fault models, Iterative logic arrays, Reversible circuits, Testability, Universal test sets.*

1. INTRODUCTION

Reversible circuits have the property that every distinct input pattern yields a distinct output pattern, and the input and output wires are equal in number. Such circuits are of interest for several reasons. Reversible circuits are information-loss less and hence tend to dissipate relatively little energy. According to Landauer's principle [1, 2], it is possible to construct a computer using reversible circuits that can compute with arbitrarily small amounts of energy. While the energy loss implied by irreversibility is orders of magnitude less than that due to other causes in current electronic technologies (e.g. high frequency of switching), this may change as VLSI feature sizes decrease and newer (nano) technologies appear. Reversibility is an essential property of the circuits needed for quantum computation [3], which is a major motivation behind this study. Certain very hard problems, notably prime factorization, can be solved using quantum algorithms that are exponentially faster than the best-known conventional (classical) methods. Reversible circuits also find application in specialized fields such as cryptography.

In this paper we study the testing properties of an important subclass of reversible circuits constructed from a library of reversible gates called controlled-NOT (CNOT) gates. We assume that all signals are bits rather than more general quantum bits (qubits). A *k-CNOT gate* has $k + 1$ input wires and $k + 1$ output wires. It transmits the first k input signals unchanged, and inverts the last input signal iff the first k inputs are all 1; clearly this input-output mapping is reversible. Figure 1 shows examples of k-CNOTs and the standard graphic symbols used for them. If $k = 0$, a k-CNOT is a simple NOT gate or inverter. 1-CNOT and 2-CNOT gates are also known as CNOT and Toffoli gates, respectively. An n-wire reversible circuit of the type of interest here—*a k-CNOT circuit*—can be constructed by starting with n wires, and iteratively concatenating k-CNOT gates to a subset of output wires of the previous circuit; see, for example, Fig. 2. It gives rise to the notion of levels in a reversible circuit. The inputs to the circuit are at level 0 and the outputs of any gate are at one plus the highest level of any of its inputs. Since a Toffoli gate can implement the NAND function, any Boolean function can be implemented in reversible fashion by a k-CNOT circuit.

Little research has been reported on testing reversible circuits. Bertrand et al. [4] have used reversibility to synthesize on-line test structures for irreversible circuits. More recently Patel et al. [5] have considered the problem of generating efficient test sets for these circuits under both the stuck-at and cell fault models. Though this problem is hard for conventional irreversible circuits, it can be much easier in the reversible case. We examine the testability of k-CNOT circuits for most commonly used (refer to [5]) single and multiple stuck-at faults. The paper is organized as follows. In Section 2, we define the notion of a universal test set (UTS), and construct UTSs for some classes of k-CNOT circuits under the SSF and MSF model. We then present

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a design method that drastically reduces UTS size for MSFs. In Section 3 by realizing a k-CNOT gate as a k-input AND gate and a 2-input EXOR gate we derive a UTS for reversible k-CNOT circuits under the SSF model. Section 4 discusses the testing of reversible iterative logic arrays (RILAs) and presents a test set generation algorithm for a d-dimensional RILA under the SCF model [10]. Section 5 concludes the paper.

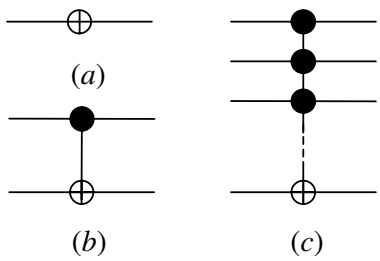

Figure 1: (a) NOT, (b) CNOT, (c) k-CNOT gate.

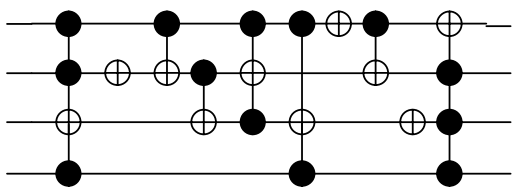

Figure 2: Example of a 4-wire k-CNOT circuit.

2. UNIVERSAL TEST SET

Let L denote a reversible gate library of interest. We define a (*uniform*) *family* of reversible circuits on L, denoted by $\boldsymbol{C}(n, L)$ or simply $\boldsymbol{C}(n)$, as all n-wire reversible circuits constructible from L for $n = 1,2,3\ldots$. A test set T is said to be *complete* for a fault model F, if it can detect all possible F-type faults in the circuit. A *universal test set* (UTS) for circuit family $\boldsymbol{C}(n)$ with respect to fault model F is a family of test sets $\boldsymbol{T}_{\text{UTS}}(\boldsymbol{C}(n))$ such that for each $\boldsymbol{C}(n)$ there is a unique test set $T(n)$ in $\boldsymbol{T}_{\text{UTS}}(\boldsymbol{C}(n))$ that is complete with respect to F for every circuit in $\boldsymbol{C}(n)$.

Theorem 1 [5]**:** A test set T is complete for a circuit C under the single and the multiple stuck-at fault models if and only if each wire in every level in C can be set to both 0 and 1 by T.

For example, if $L = \{NOT\}$, then $\boldsymbol{C}_{\text{NOT}}(n)$ is the family of reversible circuits on n wires constructed entirely from NOT gates. It is easy to see that a UTS for stuck-at faults in $\boldsymbol{C}_{\text{NOT}}(n)$ exists, in which $T(n)$ is the pair of n-bit vectors $0_n = 00\ldots0$ and $1_n = 11\ldots1$. In other words, $\boldsymbol{T}_{\text{UTS}}(\boldsymbol{C}_{\text{NOT}}(n)) = \{0_n, 1_n: n = 1, 2, 3,\}$. A trivial UTS with respect to any fault model has each $T(n)$ consisting of all 2^n possible n-bit vectors. $\boldsymbol{T}_{UTS}(C_{\text{NOT}}(n))$ as defined above consists of test sets of the minimum possible size and so is referred to as *optimal*.

Theorem 2: All stuck-at faults in an n-wire k-CNOT circuit $C(n)$ with $k \geq 2$ can be detected using at most n test vectors. $C(n)$ has a UTS of size n for MSFs.

Proof: A circuit *fixes* a test vector t if it maps input t onto output t. Every k-CNOT gate with $k \geq 2$ fixes a vector 2^i that sets just one input i to 1 and the remaining $k - 1$ inputs to 0; this is sometimes called a *weight*-1 *vector*. Thus $C(n)$ fixes all its n weight-1 vectors. The vector 2^i sets all levels of the $(i + 1)$-st wire to 1 and all other wires to 0, hence n weight-1 vectors set all levels of all wires to both 0 and 1. By Theorem 1, $\{2^i: i = 0,1,2,\ldots,n - 1\}$ is a UTS for $C(n)$.

Corollary 1: If NOT and CNOT gates are constructed from Toffoli gates, then every n-wire k-CNOT circuit with $k \geq 0$ can be extended to an $(n + 2)$-wire k-CNOT circuit that has a UTS of size $n + 2$ for MSFs.

Theorem 3: Let an n-wire k-CNOT circuit be modified by introducing an extra wire that feeds a new control input added to every k-CNOT gate. The resulting circuit has a UTS of size 3 for MSFs.

Proof: Let c denote the extra wire connected to each k-CNOT gate, which is thereby converted to a $(k + 1)$-CNOT gate. With c = 1, each $(k + 1)$-CNOT behaves like a k-CNOT, and the circuit is in operational mode. When $c = 0$, the circuit is in test mode, and every gate fixes its input signals. In the latter case, the 0_n and 1_n vectors set all n original wires on all levels to 0 and 1, respectively. All levels of the extra wire c are set to 0 by these two test vectors. A third test t that places logic 1 on c and n don't cares on all other inputs sets all levels of c to 1. The 3-member test set $\{0_n, 1_n, t\}$ constitutes a UTS by Theorem 1.

Barenco et al. [6] show how to decompose a k-CNOT gate with $k \geq 3$ into Toffoli gates. This is useful if we want to synthesize a reversible circuit from a gate library where $k \leq 2$, for example, the NCT library {NOT,CNOT,Toffoli} [7]. Next we investigate whether the above testing strategy holds after decomposition, and show that if we convert k-CNOT gates to Toffoli gates in a certain manner, a UTS of size 3 still exists.

Theorem 4 [6]: For $n \geq 5$, an $(n - 2)$-CNOT gate can be simulated by a circuit consisting of two m-CNOT gates and two $(n - m - 1)$-CNOT gates, for $2 \leq m \leq n - 3$, and one extra wire c; see Fig. 3.

Figure 4 gives an algorithm to decompose an arbitrary k-CNOT gate into Toffoli gates by repeatedly invoking Theorem 4.Finally, we observe that by combining Alg. 1 with the design modification method stated in Theorem 3, we can generate a Toffoli-gate implementation of a given n-wire k-CNOT circuit that has a UTS of size 3 for MSFs. The resulting circuit is an $(n + 2)$-wire k-CNOT circuit with two extra wires c_1 and c_2. When c_1

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and c_2 are set to 0, n original wires fix all-0 and all-1 vectors applied on them. See Fig. 5. Please note that the k-CNOT gates which operate on c_1 and c_2 receive at least one control input from n original wires. Hence, 1s at c_1 ,c_2 and 0s at other inputs sets all levels of c_1 and c_2 to 1. These 3 test vectors set all levels of all $n + 2$ wires to both 1 and 0; from Theorem 1 they constitute a UTS.

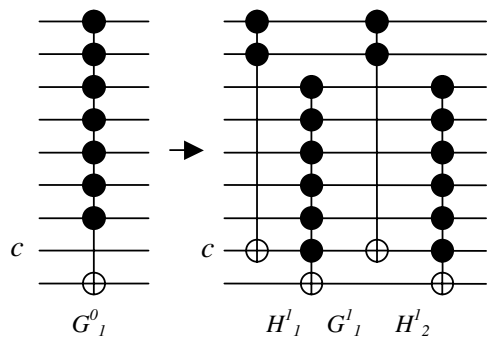

Figure 3: Illustration of Theorem 4 for $n = 9$ and $m = 2$.

Given a k-CNOT gate G^0_1 with $k \geq 3$. Introduce an extra wire c_2. Let c_1 be any control input of G^0_1.
$h = 2$; /* h is a control wire index */
for ($i = 1$; $i \leq k-2$; i++) {
if ($h == 2$) {
$h = 1$; /* Extra wire for the next step is c_1 */
for ($j = 1$; $j \leq 2^{i-1}$; j++) {
Apply Theorem 4 to $(k-i+1)$-CNOT gate G^{i-1}_j with $m = 2$ and $c = c_2$. Label the resulting two Toffoli gates H^i_{2j-1}, H^i_{2j-1} and two $2(k-i)$-CNOTs G^i_{2j-1} ,G^i_{2j}, as in Fig. 3.
} } **else** {
$h = 2$; /* Extra wire for the next step is c_2 */
for ($j = 1$; $j \leq 2^{i-1}$; j++) {
Apply Theorem 4 to $(k-i+1)$-CNOT gate G^{i-1}_j with $m = 2$ and $c = c_1$. Label the resulting two Toffoli gates H^i_{2j-1}, H^i_{2j-1} and two $2(k-i)$-CNOTs G^i_{2j-1} ,G^i_{2j}, as in Fig. 3.} }
$i = i + 1$;
}
No k-CNOT with $k > 2$ remains and the final circuit contains $3.2^{k-2} - 2$ Toffoli gates.

Figure 4: Algorithm 1 to decompose a k-CNOT gate into Toffoli gates.

The number of Toffoli gates i.e. $3.2^{k-2} - 2$ needed to simulate a k-CNOT gate can actually be reduced further. Let us analyze the case where each k-CNOT gate is optimally decomposed. Theorem 1 should now be applied at each step of decomposition with $m= p_1$, $n-m-1=p_2$, where if $(n-2)$ is odd and $n-2=2p+1$, then $p_1=p_2=p+1$; else when $(n-2)$ is even and $n-2=2p$, then $p_1=p+1, p_2=p$. To ensure that there still exists a non-trivial UTS of optimal size, i.e., 3 we need to add one more extra wire to be used as "scratch" wire in decomposition steps. To decompose a k-CNOT gate if m is chosen as mid-point as above at each step of decomposition, then number of Toffoli gates needed to stimulate a k-CNOT gate can be reduced from $(3.2^{k-2} - 2)$ to $4^{(\lceil \log_2 k \rceil)}$ (i.e. $O(k^2)$). We illustrate the new decomposition strategy for 5-CNOT gate in Fig. 6, where 16 Toffoli gates can simulate a 5-CNOT gate.

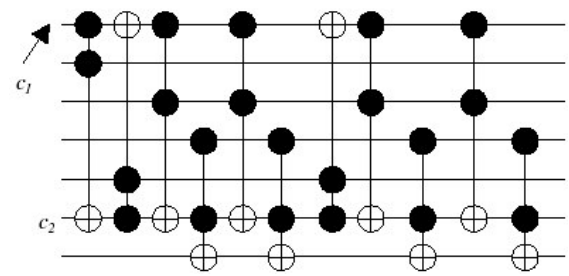

Figure 5: A circuit simulating 5-CNOT with Toffoli gates. Only half of the gates (i.e. $(3.2^{5-2} - 2)/2=11$) are shown.

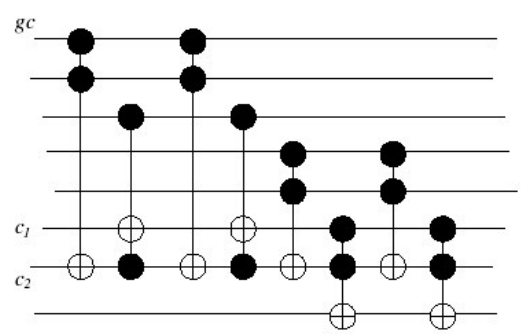

Figure 6: A circuit simulating 5-CNOT with Toffoli gates. The size of UTS remains 3. Only half of the gates (i.e. 16/2=8) are shown.

3. AND-EXOR REALIZATION OF *k*-CNOT GATES

An inadequacy of test sets based on the stuck-at fault model is that they need not cause a k-CNOT gate to invert its output, which requires that all its control inputs be set to 1 simultaneously. Thus the gate's key function (controlled inversion) is not being tested. If we model each k CNOT gate with its functional equivalent i.e. a k-input AND gate and an EXOR, any single functional gate failure can be attributed to single s-a-fs in the inputs or outputs of these gates. The main motivation behind adopting single s-a-f is that it is equivalent to single gate failure in this case. Besides, it covers majority of the multiple s-a-fs in the circuit. There are many possible realizations of EXOR gate. Besides s-a-fs on its 2 inputs and 1 output, we also consider that s-a-fs can also occur inside the EXOR gate. Irrespective of its realization all

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these faults can be covered under single s-a-f model if we test each EXOR gate exhaustively with all 4 possible input combinations. To test a k-input AND gate for s-a-fs on its inputs and output, we need to feed all-1 and k walking zero vectors [9]. We now derive a UTS for n-wire reversible circuit, where we make sure each EXOR gate is tested exhaustively, and each k-input AND gate receives $k + 1$ vectors as above.

For an n-wire reversible circuit, we propose to add n external control wires. For the k-CNOT gates with EXOR gate on the i-th wire in the original circuit, the i-th control wire feeds one new control input to each of those k-CNOT gates. k-CNOT gates necessarily become $k + 1$-CNOT gates (k-input AND gates are replaced with $(k + 1)$-input AND gates with the extra input from the i-th control wire). There are two exceptions to the functional equivalency of k-CNOT gate with k-input AND gate and EXOR gate when $k = 0,1$. In the first case, it is a simple NOT gate. When $k = 1$, there is no AND gate. The proposed DFT construction necessarily removes the degeneracy arising due to NOT gates (it becomes simple EXOR). If all the control wires are set to 1, the circuit is in operational mode. By setting the i-th control wire to 0, we can make every level of the i-th wire in the original circuit directly accessible from inputs (i.e. the i-th wire bypasses signal from its input to output). Consider the i-th wire in the original circuit again. If we set all the control wires except the i-th control wire to 0 and i-th control wire to 1, then the i-th original wire closely resembles a ***Positive Polarity Reed-Muller (PPRM)*** network of n variables (n -1 inputs come from the original wires except the i-th wire, and one input comes from i-th control wire). It has been shown that there is a UTS of size $(n + 4)$ [9] for an n-variable PPRM network under the assumption of single s-a-f model. In PPRM expression, there is at most one product term for each subset of the variables $x_1, x_2, x_3,.....,x_n$ plus a constant term C_0. It is not strictly PPRM network because it may have some repeated product terms with the same set of variables. It has been referred to as ***Positive Polarity AND-EXOR*** network in [8]. The interesting fact is that it has same testing properties as of PPRM. Hence, Reddy's testing method (4 test vectors and n walking zero vectors) [9] can be applied here with some optimizations. We assume that we feed inputs to n original wires and n control wires, and observe outputs from all of these $2n$ wires.

Consider the circuit in Fig. 2. We set all the control wires except c_1 to 0. That means the test pattern we feed to the wires except x_1 remain unchanged at all levels of those wires. Thus, the wire x_1 realizes 4-variable Positive Polarity AND-EXOR expression. Let us consider the test set $T^1 = \{T^1_1, T^1_2, T^1_3, T^1_4\}$.

	x_1	x_2	x_3	x_4	c_1	c_2	c_3	c_4
$T^1_1 \Rightarrow$	0	0	0	0	1	0	0	0
$T^1_2 \Rightarrow$	0	1	1	1	1	0	0	0
$T^1_3 \Rightarrow$	1	0	0	0	1	0	0	0
$T^1_4 \Rightarrow$	1	1	1	1	1	0	0	0

These 4 tests apply all possible combinations namely (0,0),(0,1),(1,0),(1,1) to each EXOR gate in cascade on x_1. It not only detects stuck-at faults on the wire x_1, but it also covers any single stuck-at fault inside any EXOR gate. Consider the AND gates whose outputs are fed into the EXOR gates on wire x_1 as input. T^1_4 also detects stuck-at 0 on all such AND gate outputs and AND gate inputs. To detect stuck-at 1 on AND gate outputs and AND gate inputs of 4-wire reversible circuit we need 4 walking-zero vectors, where 4 variables are x_is for $2 \le i \le 4$, and c_1. x_1 can be don't care in this case. Hence, total no. of tests needed for one wire is 4 + 4=8. So, Total number of tests needed for n wire reversible circuit is loosely $n(n + 4) = n^2 + 4n = O(n^2)$. This can be actually improved. The test T^1_1 can be done in parallel for all the wires x_is by setting all controls to 0 instead of setting control c_1 to 1 and others to 0. Out of the n walking zero vectors, consider the test vector when $c_1 = 0$, $x_i = 1$ for $2 \le i \le n$, $c_i = 0$ for $2 \le i \le n$, and x_1 ="-". Instead of applying this particular walking-zero vector for each wire separately, we can apply a test vector of the form $x_i = 1$ for $1 \le i \le n$, $c_i = 0$ for $1 \le i \le n$. So, the new bound is $n((n - 1) + 3) + 1 + 1 = n^2 + 2n + 2$; but still $O(n^2)$.

Theorem 5:
All single stuck-at faults in an n-wire reversible circuit composed of k-CNOT gates where each k-CNOT gate is modeled as a k-input AND and a 2 input EXOR gate can be detected using a UTS of size $n^2 + 2n + 2$.

4. ITERATIVE LOGIC ARRAYS

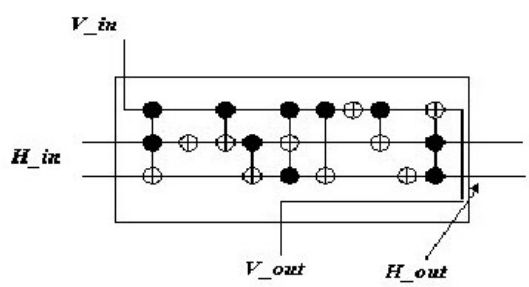

Figure 7: A Reversible Cell with 2 h-wires and 1 v-wire.

A system that consists of identical modules arranged in a geometrically regular interconnection pattern is defined as *Iterative Logic Array (ILA)*. ILAs are well-known to be easily testable. In this section we investigate the testability properties of its reversible counterpart i.e. *reversible Iterative Logic Array (RILA)*. Each single module in ILA is referred to as cell. A cell, which receives h horizontal inputs h_{in} s, v vertical inputs v_{in} s

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and produces h horizontal outputs h_{out} s, v vertical outputs v_{out} s and implements its internal logic by k-CNOT gates with $k \leq h+v-1$ is also *reversible* i.e. it maps every distinct input (h_{in}, v_{in}) to a distinct output (h_{out}, v_{out}). It is referred as *(h, v) reversible cell.* See Fig. 7. We denote the cell function realized by *(h, v) reversible* cells f i.e. f: $\{0,1\}^h \times \{0,1\}^v \Rightarrow \{0,1\}^h \times \{0,1\}^v$. f is clearly ***bijective***. A system that consists of reversible cells arranged in a geometrically regular interconnection pattern is defined as *reversible Iterative Logic Array (RILA)***.** As all cells are identical it is also called "*homogeneous*". Please note that *(h,v)* reversible cells can be arranged to form RILA of maximum 2 dimensions.

Theorem 6[10]**:**
All SCFs in a *one*-dimensional homogeneous RILA of p *(h,v)*-cells can be detected with 2^{h+v} test vectors independent of p.

Proof: As the cell function f realized by (h,v) reversible cell is *bijective*, any single cell fault propagates to either h_{out} or v_{out} of the cell in which the failure occurs. As all other cells are bijective, this ensures that any SCF propagates to an observable output (h_{out} of the rightmost cell or v_{out}s of all p cells) for any vertical input v_{in} to that faulty cell. This implies 2^{h+v} tests suffice to detect all SCFs.

Given *d*-Dimensional Homogeneous Reversible ILA *(RILA)* with dimensions $\{k_1, k_2, k_3, \ldots\ldots, k_d\}$.
Select the dimension k_1.
Initialize *TS=NULL*; TS_{new} = *NULL*;
Construct EFD of degree 1 (x-transition graph) for the 1-dimensional RILA with the dimension k_1.
Decompose the EFD in all possible ways to generate all possible test sets ***ts*** s and include ***ts*** in ***TS***;
for(i =2; $i \leq$ d; i ++){
for each test set ***ts*** in ***TS***{
Construct EFD of degree i with the test set ***ts*** and dimension k_i.
If (the EFD can be decomposed into disjoint Eulerian sub graphs){
Decompose the EFD of degree i in all possible disjoint Euler tours.
For each decomposition generate a test set ts_{new}.
Include ts_{new} in TS_{new}.
} /* end if */
} /* end for each test set .. */
$TS = TS_{new}$.
TS_{new} = *NULL;*
}
output ***TS***. /* each test set ***ts*** in ***TS*** is a solution */

Figure 8: Test Generation Algorithm for *d*-dim RILA

2^{h+v}-testability implies that the *x-transition graph* of any one-dimensional Reversible ILA consists of disjoint Eulerian sub graphs [10] (ex. Fig. 9(a)). We refer to this graph as *Effective Functional Diagram* (EFD) of degree 1. Besides, 2^{h+v}-testability imposes that cell function f is also ***injective*** i.e. for a fixed vertical input; horizontal output of a reversible cell is a permutation of horizontal inputs. It can be shown that all SCFs in *one*-dimensional RILA can be detected using constant number of test vectors independent of p by observing only the h_{out}s of the rightmost cell, provided any SCF propagates to the horizontal output of the faulty cell [11]. The idea of (h,v) reversible cells can actually be extended to $(k_1,k_2,k_3,\ldots\ldots, k_d)$ reversible cells which can then be interconnected to form RILA of maximum d dimensions. The cell function f (f: $\{0,1\}^{k_1} \times \{0,1\}^{k_2} \times \ldots \times \{0,1\}^{k_d} \Rightarrow \{0,1\}^{k_1} \times \{0,1\}^{k_2} \times \ldots \times \{0,1\}^{k_d}$) remains *bijective* as before. Using similar argument it can be proven that all SCFs in any d-dimensional RILA can be detected by $2^{\{k_1+k_2+\ldots+k_d\}}$ test vectors independent of number of cells.

Theorem 7: All SCFs in any d-dimensional homogeneous RILA of $(k_1,k_2,k_3,\ldots\ldots, k_d)$-cells can be detected with $2^{\{k_1+k_2+\ldots+k_d\}}$ test vectors independent of number of cells.

Let a d-dimensional RILA consist of dimensions from the set S_d $\{k_1, k_2, k_3,\ldots, k_d\}$. Consider 2 subsets S_p and S_{p+1} of cardinalities p and $p+1$ from this set such that $2 \leq p+1 \leq d$ and $S_p \subset S_{p+1} \subseteq S_d$. The p-dimensional RILA with dimensions from the set S_p is $2^{\{k_1+k_2+\ldots+k_d\}}$ –testable. Let the *Effective Functional Diagram (EFD)* of degree p of this p-dimensional RILA consists of disjoint Eulerian sub-graphs. All possible test sets ***ts*** s each of size $2^{\{k_1+k_2+\ldots+k_d\}}$ are derived by decomposing the EFD of degree p in all possible ways. It can be shown that there is at least one test set ***ts*** for which the EFD of degree $p+1$ constructed for the $(p+1)$-dimensional RILA with dimensions from the set S_{p+1} also consists of disjoint Eulerian sub-graphs [10]. Based on this property we now present a test set generation algorithm for any d-dimensional RILA in Fig. 8. We show the test set generation for 2D RILA constructed from the cells of Fig. 7 using this algorithm in Fig. 9.

The x-transition diagram of Fig. 9(a) (constructed for the (2,1) reversible cell of Fig. 7) can be decomposed into disjoint Eulerian sub-graphs in 2 ways generating 2 test sets TS^1 and TS^2 each of size $2^3 = 8$. To generate TS^1 we consider 2 cycles i.e. $\{(q_2, 0)\text{->}(q_3, 0)\}$ and $\{(q_1, 1)\text{->}(q_3, 1)\}$ in the x-transition graph. $TS^1_1 \Rightarrow$ applies $(q_0, 0)$ in all cells, $TS^1_2 \Rightarrow$ applies $(q_0, 1)$ in all cells, $TS^1_3 \Rightarrow$ applies $(q_1, 0)$ in all cells, $TS^1_4 \Rightarrow$ applies $(q_2, 1)$ in all cells, $TS^1_5 \Rightarrow$ $(q_2, 0)$ in odd cells and $(q_3, 0)$ in even cells, $TS^1_6 \Rightarrow$ $(q_3, 0)$ in odd cells and $(q_2, 0)$ in even cells, $TS^1_7 \Rightarrow$ $(q_1, 1)$ in odd cells and $(q_3, 1)$ in even cells, $TS^1_8 \Rightarrow$ $(q_3, 1)$ in odd cells and $(q_1, 1)$ in even cells. The EFD of degree 2 constructed for the 2D RILA using the test set TS^1 is shown in Fig. 9(b). It is not clearly decomposable into disjoint Euler tours. We consider instead a single cycle $\{(q_2, 0)\text{->}(q_3, 1)\text{->}(q_1, 1)\text{->}(q_3, 0)\}$ in the x-transition

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diagram of Fig. 9(a) to generate TS^2. First 4 test vectors of TS^2 are similar to those of TS^1.

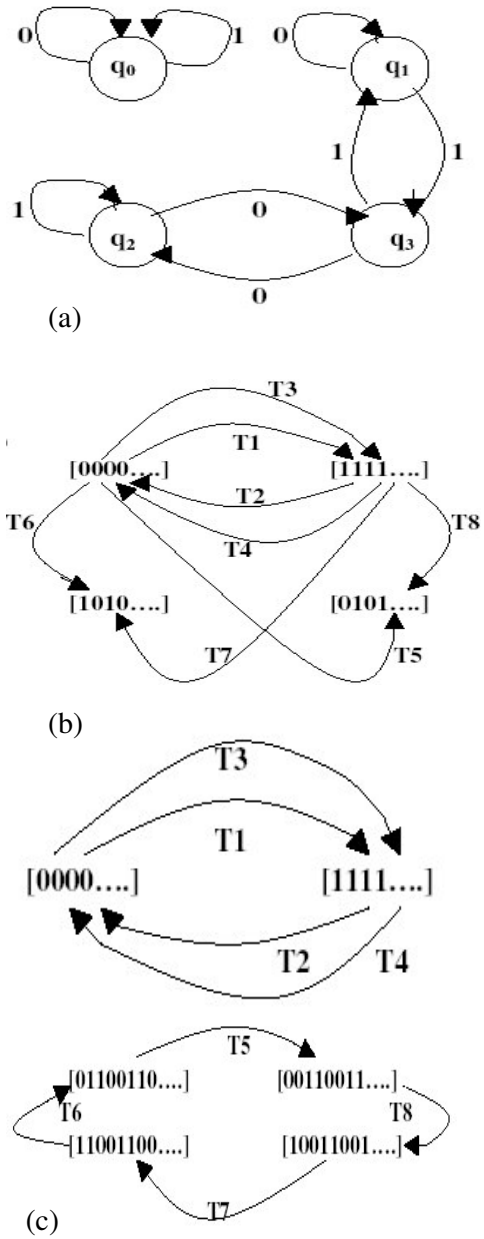

Figure. 9: ***(a)*** x-transition diagram (EFD of degree 1) for the cell state table in (a). ***(b)*** EFD of degree 2 when 2 cycles are considered. ***(c)*** EFD of degree when a single cycle is considered.

$TS^2_5 \Rightarrow$ (q_2, 0) in cells 1,5,9,13… (q_3,1) in cells 2,6,10,14… (q_1, 1) in cells 3,7,11,15… (q_3, 0) in cells 4,8,12,16…
$TS^2_6 \Rightarrow$ (q_3, 1) in cells 1,5,9,13… (q_1,1) in cells 2,6,10,14… (q_3, 0) in cells 3,7,11,15… (q_2, 0) in cells 4,8,12,16…
$TS^2_7 \Rightarrow$ (q_1, 1) in cells 1,5,9,13… (q_3, 0) in cells 2,6,10,14… (q_2, 0) in cells 3,7,11,15… (q_3, 1) in cells 4,8,12,16…
$TS^2_8 \Rightarrow$ (q_3, 0) in cells 1,5,9,13… (q_2, 0) in cells 2,6,10,14… (q_3, 1) in cells 3,7,11,15… (q_1, 1) in cells 4,8,12,16…

The EFD of degree 2 using the test set TS^2 (shown in Fig. 9(c)) is clearly decomposable into disjoint Eulerian sub-graphs and gives a solution for a test set of size 8 to detect all SCFs in the 2D RILA. The test set TS^{2D} is described below:

$TS^{2D}_1 \Rightarrow$ applies TS^2_1 in odd rows, TS^2_2 in even rows; $TS^{2D}_2 \Rightarrow$ applies TS^2_2 in odd rows, TS^2_1 in even rows; $TS^{2D}_3 \Rightarrow$ applies TS^2_3 in odd rows, TS^2_4 in even rows; $TS^{2D}_4 \Rightarrow$ applies TS^2_4 in odd rows, TS^2_3 in even rows. TS^{2D}_5, TS^{2D}_6, TS^{2D}_7, TS^{2D}_8 alternate TS^2_6, TS^2_5, TS^2_8, TS^2_7 in every 4 rows of the 2D RILA.

5. CONCLUSION

In this paper we have derived Universal Test Sets for a very important subclass of reversible circuits that are composed of *k*-CNOT gates. Though reversibility simplifies testing in general, it is noted that adding extra test structures to a reversible circuit such that the modified circuit remains reversible is difficult. The Iterative Logic Arrays possess some interesting testability properties; we have shown that its reversible counterpart is also easily testable.

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